Cedric Phillips

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Montreal, QC, Canada

Education

- **McGill University** — *Fall 2022 - Winter 2026 (expected)*
  B.Sc. Honours Applied Mathematics — Montreal, QC
  Minor concentrations: GIS & Remote Sensing, Hindi Language
  - GPA: 3.60/4.00
    Winter 2025: Exchange at the École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
  - Coursework: Graduate probability theory & mathematical stats., numerical analysis, stochastic processes.

Experience

- **McGill University** — *Winter 2025*
  Grader, MATH 458: Honours Differential Geometry — Funded by TEAM
  - Verified solutions to assigned problems and returned assignments to all students with detailed feedback in a consistent, timely manner as the sole grader for the course

Technical Skills & Proficiencies

- **Programming/data analysis:** Python (polars, NumPy, matplotlib), R (qrmtools), SQL, Java
- **Mathematical:** LaTeX, Mathematica, Lean prover
- **Languages:** English (native), French (native), Hindi-Urdu (advanced)

Projects

- **Honours Research Project** — *Winter 2026*
  Hierarchical copula dependence models — Supervised by Prof. Nešlehová
  - Developing ways to infer the tree structure of a hierarchical model for multivariate Bernoulli variables from Kendall and Spearman rank correlation matrix data
  - Generating and transforming samples from the model using R and investigating applications of the model to credit risk, culminating in a final report

- **Stochastic optimization in asymmetric games** — *Present-April 2026*
  Maximal scoring strategies for the Lizard Cult in Root — To appear in McGill's undergraduate math journal
  - Created a Markov chain model for the Root game state and the Lizard Cult's play options; analyzed decision-making under uncertainty variables to find moves that optimize expected value
  - Validated results of project via Monte Carlo simulation, produced visualizations of scoring trajectories with parameter sensitivity using Python

- **Directed Reading Project** — *Summer 2024*
  Proof formalization in the Lean language — Supervised by Dr. Turcotte
  - Under guidance of a McGill PhD. student, members of our project group studied the underlying theory and use of the Lean language from scratch
  - Explored areas of personal mathematical interest, learning advanced proof techniques as necessary, culminating in an original constructive Lean-verification of the Cauchy-Schwarz inequality in $\mathbb{R}^n$

- **Nonlinear Dynamics Honours Project** — *Fall 2023*
  Canards and other explosive behaviours — Supervised by Prof. Humphries
  - Independently conducted research on a chosen graduate-level topic: canard 'explosions', Hopf bifurcations, and the 'blow-up' map in slow-fast dynamical systems
  - Read literature to produce a referenced report summarizing the most important insights & applications

Work

- **Wild Blue Restaurant & Bar** — *Summers and winters, 2024-2026*
  Expeditor — Whistler, BC
  - Logistics under pressure: quality-checked all dishes being sent out of the kitchen, organized a non-stop flow of orders & the resources needed to send them out

- **Caramba Restaurant** — *Summers 2021-2024*
  Busser & Expeditor — Whistler, BC